



# Associations of *BDNF* methylation and rs6265 genotype with cognitive response to a six-month aerobic exercise intervention in women with early-stage breast cancer: results from the EPICC Trial

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Received: 26 June 2025 / Accepted: 30 October 2025 / Published online: 18 November 2025  
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**Abstract** Women with breast cancer often experience cognitive decline, an accelerated aging phenotype. While aerobic exercise may mitigate this decline, effects are inconsistent by domains, and the molecular mechanisms remain unknown. Given the brain-derived neurotrophic factor (*BDNF*)’s responsiveness to exercise and role in cognition, we investigated *BDNF* methylation and rs6265, its functional SNP, with cognitive responses to aerobic exercise. Leveraging data from a randomized clinical trial which found cognitive function improved with a six-month aerobic exercise than control group in women with breast cancer, we included sub-samples with either pre-randomization or post-intervention M-values. CpG-site level M-values (higher positive value=greater

methylation) of *BDNF*, rs6265 genotype (CC/CT/TT), composite scores for each cognitive domain (higher scores=better performance), and linear mixed-effect modeling were used. Women ( $N=117$ , 75% of trial participants) were on average  $62.6 \pm 7.84$  years old. The majority were White (89.7%). Intervention effects were observed at four CpG sites: cg05189570, cg08382738, cg12067298, cg20340655. Methylation increases at cg05818894 ( $b=0.138$ ;  $p=0.049$ ) and cg06025631 ( $b=0.373$ ;  $p=0.031$ ) were correlated with attention and mental flexibility improvements, respectively. Working memory improvement in the exercise group was greater with less methylation increases at cg12296752 and cg15462887, but with greater methylation increases at cg06025631, cg04481212, and cg16257091. Processing speed improvement in the exercise group was greater with greater methylation increase at cg06260077. At pre-randomization, the additive T allele effect of rs6265 on

**Supplementary Information** The online version contains supplementary material available at <https://doi.org/10.1007/s11357-025-02004-x>.

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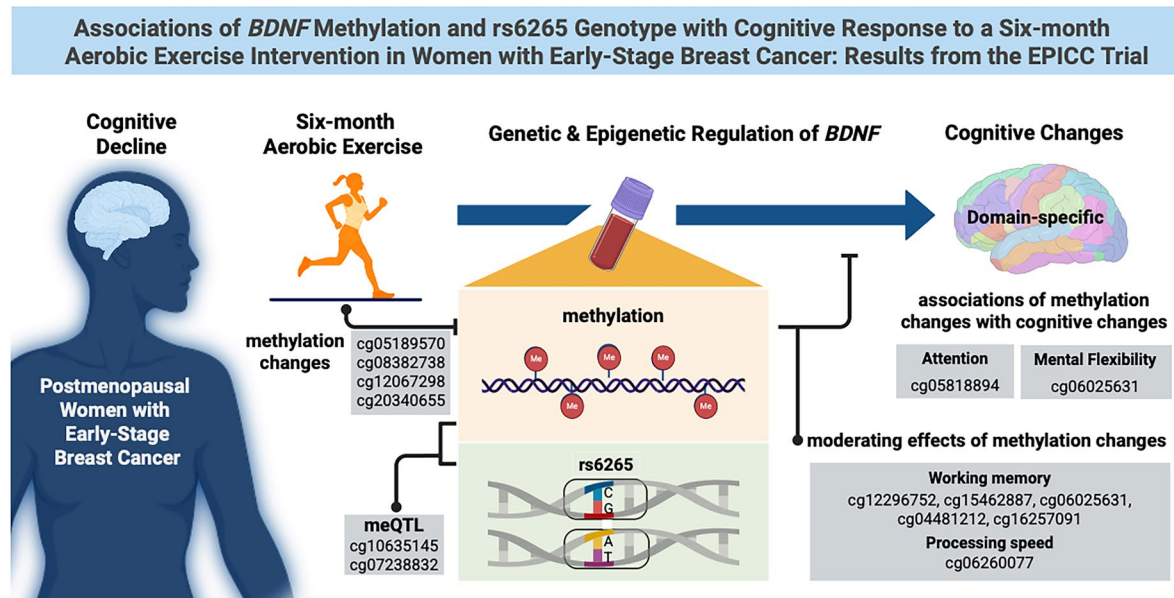
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methylation of cg10635145 and cg07238832 was detected. Findings suggest aerobic exercise may exert cognitive benefits through epigenetically regulated mechanisms,

highlighting CpG-specific methylation as potential targets for managing cognitive decline in women with breast cancer.

**Graphical Abstract** Created in BioRender. Cho, M. (2025) <https://BioRender.com/931pay3>



**Keywords** Brain-derived neurotrophic factor · Cognitive aging · Cognition · DNA Methylation · Aerobic exercise · Rs6265 genotype

## Introduction

Normal aging processes can be accelerated in patients with cancer following cancer and cancer treatment, which leads to accelerated aging phenotypes such as cognitive decline [1–4]. Cognitive decline, reported among up to 75% of women with breast cancer, may begin before or during cancer treatment [5] and persist for years after treatment completion [6], reducing the ability to meet daily demands and associated with a deteriorating quality of life [7, 8].

Aerobic exercise is one promising strategy to mitigate cognitive decline with cancer and its treatment [9–11]. Physiological changes following aerobic exercise include epigenetic modifications that influence gene expression without changing the DNA sequence.

Given the critical role of DNA methylation in both cancer and the aging process [12], aerobic exercise may mitigate phenotypes of accelerated aging effects following cancer and cancer treatment. Our earlier work showed improvement in processing speed following six months of moderate-intensity aerobic exercise in women with breast cancer [13], and cross-sectional associations between *BDNF* methylation and processing speed at pre-randomization [14] from the Exercise Program in Cancer and Cognition (EPICC) randomized controlled trial (RCT). To build on these findings, we are conducting follow-up analyses using the EPICC RCT data to gain deeper insight into the potential mechanisms and moderators underlying the observed cognitive response following exercise intervention.

Brain-derived neurotrophic factor (*BDNF*) plays a role in neural cell creation, protection, and regulation, as well as synaptic plasticity [15]. Lower serum and plasma-derived *BDNF* levels are often observed in Alzheimer's disease, and other common aging-related conditions [16], and have been associated with an increased risk of cognitive decline in older populations [17] and patients with cancer [18–20]. Increases in *BDNF*

following exercise were also associated with improvements in hippocampal volume and memory [21, 22].

The rs6265 is a functional single nucleotide polymorphism in the *BDNF* gene that results in Valine (Val) being replaced by Methionine (Met) at codon 66, which alters the folding of the pro-*BDNF* protein, making it harder to transport inside the cell and release during activity [23]. There are three possible genotypes: Val66Val (CC), Val66Met (CT), and Met66Met (TT). The CC genotype was associated with better cognitive function [24]. Aerobic exercise-induced increases in *BDNF* were greater in individuals with the CT genotype than those with the CC genotype among older adults [25]. Recent systematic reviews demonstrate the crucial role of the rs6265 genotype in moderating the cognitive benefits of aerobic exercise but with differential effects by T allele presence [26, 27]. These results suggest the possibility that exercise-induced changes in *BDNF* and cognitive function could vary depending on population, rs6265 genotype, and cognitive domains.

Previous studies have shown a lot of potential benefits of aerobic exercise to cognitive function and the brain. However, gaps largely remain in the mechanisms and moderators of the effects in humans. Furthermore, integrating underlying biological mechanisms into interventions may maximize their effectiveness [28] and addressing the remaining gaps can contribute to the development of effective personalized exercise interventions to optimize cognitive outcomes.

Building on primary findings from the EPICC RCT, the current analysis explores 1) whether DNA methylation of the *BDNF* gene changes in response to six months of aerobic exercise, 2) whether pre-randomization DNA methylation differs by rs6265 genotype, 3) whether changes in methylation are associated with changes in cognitive function, and 4) whether the cognitive response to six months of aerobic exercise is moderated by the changes in methylation and/or by the rs6265 genotype among women with breast cancer. We expected CpG-site-specific methylation changes, with smaller changes in the exercise group. Moreover, we expected decreased methylation in individuals carrying the T allele of rs6265, negative associations of changes in methylation with changes in cognitive function, and differential cognitive benefits depending on the level of methylation changes and/or rs6265 genotype.

## Methods

### Design

Employing a single site 6-month RCT design, this exploratory analysis used de-identified data from the EPICC trial, which examined whether, compared to usual care, moderate-intensity aerobic exercise improved cognitive function in postmenopausal women with early-stage breast cancer receiving endocrine therapy (R01-CA196762) over 6 months, and the companion study, which investigated the role of epigenomics in the EPICC trial (R01-CA221882). The EPICC trial was registered (ClinicalTrials.gov NCT02793921) and approved by the Institutional Review Boards (IRBs) of the University of Pittsburgh (PRO15120433), St. Clair Hospital (PRO1712001), and Carnegie Mellon University (study2016\_00000197), and the current study was approved by the University of Pittsburgh IRB (STUDY25010132). Informed consent was obtained prior to data collection. Peripheral blood samples and cognitive function assessments from a total of 153 participants [exercise group ( $n=77$ ) and usual care group ( $n=76$ )] were collected at baseline (pre-randomization) and within two weeks after the completion of the intervention for the exercise group, with equivalent timing for the usual care group (post-intervention).

### Study population

Inclusion and exclusion criteria for the EPICC trial are as follows: women who were a) postmenopausal, b) younger than 80 years old, c) diagnosed with ductal carcinoma in situ, or stage I, IIa, IIb, or IIIa breast cancer, d) eligible to receive endocrine therapy, and e) English-speaking with a minimum of 8 years of education, but without a) a diagnosis of any type of cancer before breast cancer (excluding some skin cancers), b) clinical evidence of distant metastases, c) self-reported hospitalization for psychiatric illness within the past two years, d) history of neurological illness, e) breast cancer surgery complications, or f) reconstructive surgery within the study period. With the advent of the COVID-19 pandemic, the inclusion criteria were expanded to include women within two years of completing primary therapy (surgery  $\pm$  chemotherapy).

For the current analysis, participants with DNA methylation data at either pre-randomization or post-intervention were eligible, yielding a total of 117 participants (61 in the exercise group and 56 in the usual care group). Statistical power was estimated using PASS 2024 (version 24.0.2) [29]. A previous study in patients with Alzheimer's disease reported a cross-sectional association between DNA methylation at *BDNF* promoter region CpG sites and cognitive function with medium-to-large effect sizes with  $r \approx -0.65$  [30]. While these findings suggest a potentially strong link, the current longitudinal design, broader set of CpG sites, and focus on women with breast cancer prompted the use of a more conservative estimate, assuming a moderate effect size. Under this assumption, the available sample size of 117 provided sufficient power ( $>80\%$ ) to detect large effect sizes between changes in methylation and cognitive function, even after accounting for covariates. For the group-by-time interaction testing the effect of aerobic exercise on methylation changes, a two-group repeated measures design (with two time points) was modeled, assuming a slope difference of 0.5 and a standard deviation of 1. This yielded a statistical power of 0.521, indicating moderate sensitivity but limited power to detect small intervention effects.

#### EPICC trial intervention

The exercise group underwent a six-month moderate-intensity aerobic exercise intervention. Participants gradually increased their activity to reach a minimum of 150 min of moderate-intensity aerobic exercise per week. Research staff with American College of Sports Medicine certification in exercise training supervised exercise sessions at the University of Pittsburgh and community-based sites. During the COVID-19 stay-at-home restrictions, monitored remote exercise sessions were conducted via Zoom or telephone calls to replace the supervised in-person exercise sessions. The usual care group was asked to maintain their regular daily activities. More detailed information on EPICC methodology can be found in previous publications [13, 31].

#### Measures

##### *Cognitive function*

Neurocognitive tests were selected based on their sensitivity to change across multiple domains of neurocognitive function in women with breast cancer [20]. Cognitive function composite scores were computed as the average Z-scores from the identified individual neurocognitive test values for each of seven cognitive domains—attention, learning and memory, verbal memory, executive function, working memory, mental flexibility, and processing speed—after normalization based on age and education. Cognitive domains were identified by exploratory factor analysis. More detailed information regarding the derivation of cognitive function composite scores is available in previous publications [14, 19]. Higher composite scores indicate better cognitive function.

##### *DNA methylation*

Using the banked EPICC trial pre-randomization and post-intervention peripheral blood samples, the companion study collected whole genome DNA methylation data using the Illumina Infinium Methylation EPIC BeadChip (hg19). Batch effects are mitigated through functional normalization, further enhancing reliability [21, 22]. Using minfi and Enmix R packages, quality control was performed, and low-quality probes were detected and removed. M-values for *BDNF* at individual CpG site levels, including  $\pm 2$  kb 5' and 3' of the gene to capture local regulatory elements were abstracted for this analysis. A total of 88 CpG sites in the *BDNF* gene were retained after quality control.

##### *rs6265 Genotype*

The rs6265 genotyping was conducted using TaqMan® Allele discrimination technology using the Applied Biosystems™ QuantStudio™ 3 Real-Time (PCR) System. Data quality was ensured by manually reviewing the allelic discrimination plots, genotypes double-blind called and compared, and deviation from Hardy–Weinberg equilibrium assessed.

## Covariates

Covariates included age [years], education [years], and body mass index [BMI; kg/m<sup>2</sup>]. These were selected based on prior literature identifying their relevance to cognitive function and DNA methylation patterns in cancer and aging populations [32, 33]. Group assignment [0=usual care group, 1=exercise group] was included as a covariate in all statistical models to control potential group-level differences. However, in analyses where the primary objective was to examine the effect of aerobic exercise on DNA methylation, group assignment was treated as the main independent variable rather than as a covariate. Other behavioral and comorbidity factors (smoking status=1.7%, cardiac disease=9.7%, diabetes mellitus=7.4%) were evaluated but not included as covariates due to their low prevalence and limited variability in the current sample.

## Statistical analysis

Statistical analyses were conducted using R (version 4.4). Data were first screened for data abnormalities (e.g., missing values, outliers). Pre-randomization characteristics were summarized using means, standard deviations (SD), and percentages. Group differences were assessed using two-sample t-tests for continuous variables and chi-square tests for categorical variables ( $\alpha=0.05$ ). Assumptions were checked via residual analysis; diagnostic plots and influence statistics revealed no influential cases. To assess whether rs6265 genotype distribution in the current subsample of participants followed Hardy–Weinberg Equilibrium, a chi-square test was conducted with an expected frequency threshold of  $\geq 5$  per genotype category. Multicollinearity among CpG site M-values was assessed at each time point using a variance inflation factor threshold of 10. Patterns of missingness were evaluated to determine randomness. Missing M-values were addressed via multiple imputation using linear regression, with age and BMI included as auxiliary variables to improve accuracy. Five imputed datasets were generated.

To explore group-by-time effects of intervention (exercise vs usual care) on methylation, linear mixed-effect modeling with maximum likelihood estimation was performed, controlling for age and BMI. The main effects of time and group are reported following

the group-by-time effects. The standardized mean difference was used to summarize the group-by-time, time, and group effects on the changes in methylation over six months.

For methylation quantitative trait loci (meQTL) analysis, assessing cross-sectional associations of rs6265 genotype (CC vs CT vs TT) with methylation across 88 CpG sites at pre-randomization, the Matrix eQTL package (version 2.4, [34]) was used. We applied an additive genetic model, assuming that the effect of rs6265 genotype on methylation is dose-dependent, with each copy of the T allele (CT, TT) contributing proportionally to changes in methylation. Sensitivity analyses were performed by adjusting models with/without TT cases given its lower prevalence.

To explore the associations between changes in methylation as the predictor and changes in cognitive composite score as the outcome, changes in methylation and cognitive composite score were each calculated by subtracting pre-randomization values from post-intervention values, and multiple linear mixed-effect modeling was performed with and without adjustment for group assignment, age, and education. To summarize the associations between the outcomes and the predictors, adjusted unstandardized coefficients (*b*), 95% confidence interval (95%CI), and p-values were reported. For CpG sites showing significant intervention effects, subgroup analyses of the intervention group assignment were conducted to explore whether the associations of changes in DNA methylation with changes in cognitive function differed between the groups. Model fit was assessed using the Bayesian Information Criterion (BIC), comparing an unadjusted model and an adjusted model, when a lower BIC indicates better model fit.

Moderation effects of changes in methylation of each CpG site and rs6265 genotype (CC vs CT + TT) on cognitive response to aerobic exercise were assessed using the PROCESS macro (model 1; version 4.3 [35]). For this exploration, participants with complete DNA methylation data at both time points were used ( $n=64$ ). Model specifications included rs6265 genotype, changes in methylation, and group assignment, and interaction terms (group assignment  $\times$  rs6265 genotype and group assignment  $\times$  changes in methylation respectively). Bootstrap resampling with 5,000 samples was used to generate bias-corrected confidence intervals for



moderation effects, with statistical significance determined by confidence intervals excluding zero.

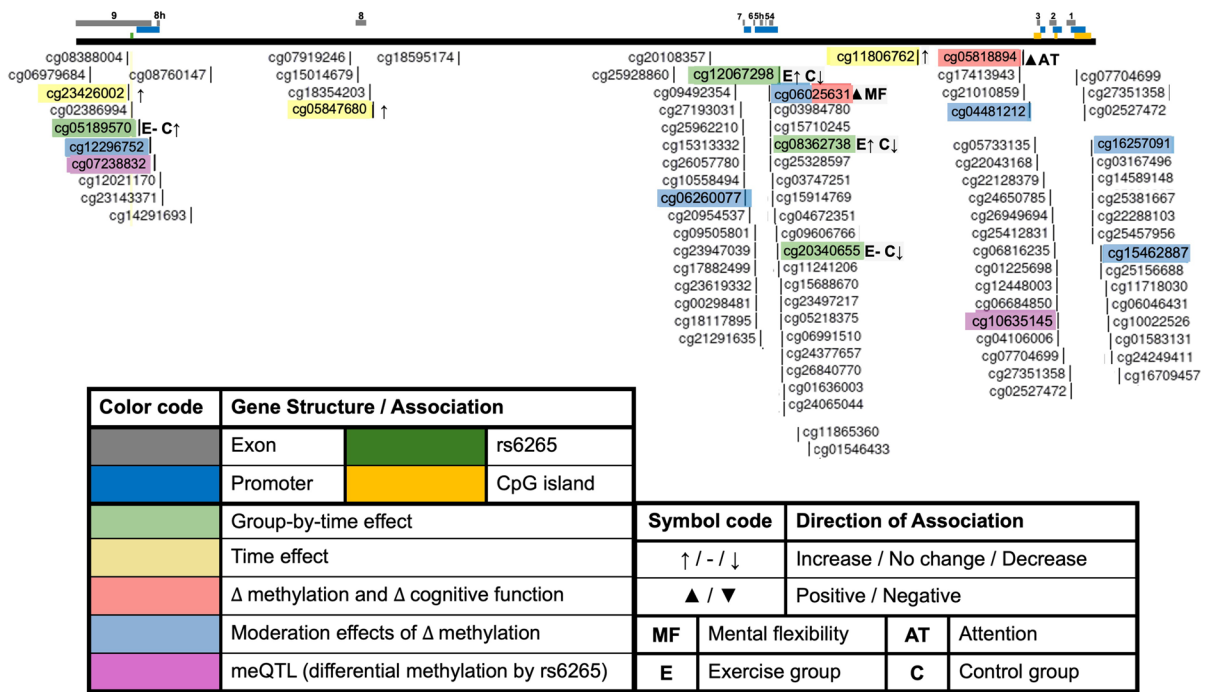
Multiple statistical tests were performed across all study objectives. For the cross-sectional meQTL analysis, Bonferroni correction (0.05/88) was applied because this objective tested a specific set of independent genotype and CpG-site methylation associations for a well-characterized functional polymorphism in the *BDNF* gene (rs6265). Given its strong prior biological relevance, stringent control of the family-wise error rate was warranted. Nonetheless, although multiple tests were also conducted for other aims, given the exploratory nature of the current analysis, we prioritized minimizing Type II errors to avoid overlooking potentially meaningful biological signals and did not correct for multiple testing. Instead, we reported unadjusted p-values and conducted sensitivity analyses for each aim (restricting to White participants and including/excluding TT genotypes) to evaluate the robustness of the findings.

Results

Figure 1 visually summarizes the findings of the current analysis and Table 1 reports pre-randomization characteristics of the participants. Participants (N = 117) were on average 62.6 years old (SD = 7.84), mostly white (89.7%) and diagnosed with stage I breast cancer (61.5%). Cognitive composite scores were comparable, except for verbal memory, which was higher in the exercise group ( $p = 0.042$ ). The rs6265 genotype distribution did not deviate from Hardy–Weinberg equilibrium ( $\chi^2 = 0.005$ ,  $p = 0.950$ ).

Intervention effects on DNA methylation

Significant group-by-time effects of intervention on DNA methylation were identified at cg05189570 ( $F_{\text{gxt}} = 8.546$ ,  $p = 0.005$ ), cg08362738 ( $F_{\text{gxt}} = 4.456$ ,  $p = 0.038$ ), cg12067298 ( $F_{\text{gxt}} = 7.133$ ,  $p = 0.009$ ), and cg20340655 ( $F_{\text{gxt}} = 7.799$ ,  $p = 0.006$ ) (Fig. 2, Supplementary Material 1). Given the modest a priori statistical power (0.521) to detect group-by-time interaction effects on DNA methylation changes, these



**Fig. 1** This figure summarizes the key findings from the current study. Associations are represented using distinct colors and symbols. The upper portion illustrates the gene structure,

including exons, rs6265, promoter regions, and CpG island. Δ indicates “change in”, and meQTL refers to methylation quantitative trait loci

**Table 1** Pre-randomization Sample Characteristics ( $n = 117$ )

|                                          | Total                | Exercise Group ( $n = 61$ ) | Control Group ( $n = 56$ ) | p      |
|------------------------------------------|----------------------|-----------------------------|----------------------------|--------|
|                                          | Mean (SD) or $n$ (%) | Mean (SD) or $n$ (%)        | Mean (SD) or $n$ (%)       |        |
| Demographic and Clinical Characteristics |                      |                             |                            |        |
| Age [years]                              | 62.6 (7.84)          | 62.2 (8.49)                 | 62.9 (7.13)                | 0.633  |
| Education [years]                        | 16.3 (2.59)          | 16.5 (2.78)                 | 16.0 (2.37)                | 0.327  |
| Race                                     |                      |                             |                            | 0.433  |
| White                                    | 105 (89.7)           | 53 (86.9)                   | 52 (92.9)                  |        |
| African American/Black                   | 8 (6.8)              | 6 (9.8)                     | 2 (3.6)                    |        |
| Other <sup>‡</sup>                       | 4 (3.5)              | 2 (3.2)                     | 2 (3.6)                    |        |
| BMI [kg/m <sup>2</sup> ]                 | 30.6 (6.45)          | 30.4 (7.14)                 | 30.9 (5.65)                | 0.638  |
| Days on endocrine therapy                | 92.3 (187.90)        | 91.1 (173.17)               | 93.7 (204.32)              | 0.941  |
| Disease stage                            |                      |                             |                            | 0.236  |
| DCIS                                     | 19 (16.2)            | 13 (21.3)                   | 6 (10.7)                   |        |
| I                                        | 72 (61.5)            | 33 (54.1)                   | 39 (69.6)                  |        |
| IIa                                      | 16 (13.7)            | 8 (13.1)                    | 8 (14.3)                   |        |
| IIb                                      | 6 (5.1)              | 5 (8.2)                     | 1 (1.8)                    |        |
| IIIa                                     | 4 (3.4)              | 2 (3.3)                     | 2 (3.6)                    |        |
| Chemotherapy                             |                      |                             |                            | 0.583  |
| Yes                                      | 19 (16.2)            | 11 (18.0)                   | 8 (14.3)                   |        |
| No                                       | 98 (83.8)            | 50 (82.0)                   | 48 (85.7)                  |        |
| Cognitive Function Composite Score       |                      |                             |                            |        |
| Learning and Memory                      | −0.1 (0.74)          | −0.1 (0.74)                 | −0.1 (0.76)                | 0.904  |
| Attention                                | −0.2 (0.70)          | −0.2 (0.76)                 | −0.2 (0.64)                | 0.822  |
| Processing Speed                         | 0.1 (0.69)           | 0.2 (0.63)                  | 0.1 (0.74)                 | 0.491  |
| Working Memory                           | −0.4 (0.79)          | −0.5 (0.72)                 | −0.3 (0.85)                | 0.327  |
| Verbal Memory                            | −0.4 (0.90)          | −0.2 (0.90)                 | −0.5 (0.88)                | 0.042* |
| Executive Function                       | 0.3 (0.64)           | 0.3 (0.65)                  | 0.2 (0.63)                 | 0.890  |
| Mental Flexibility                       | 0.1 (0.82)           | 0.1 (0.75)                  | 0.1 (0.90)                 | 0.626  |

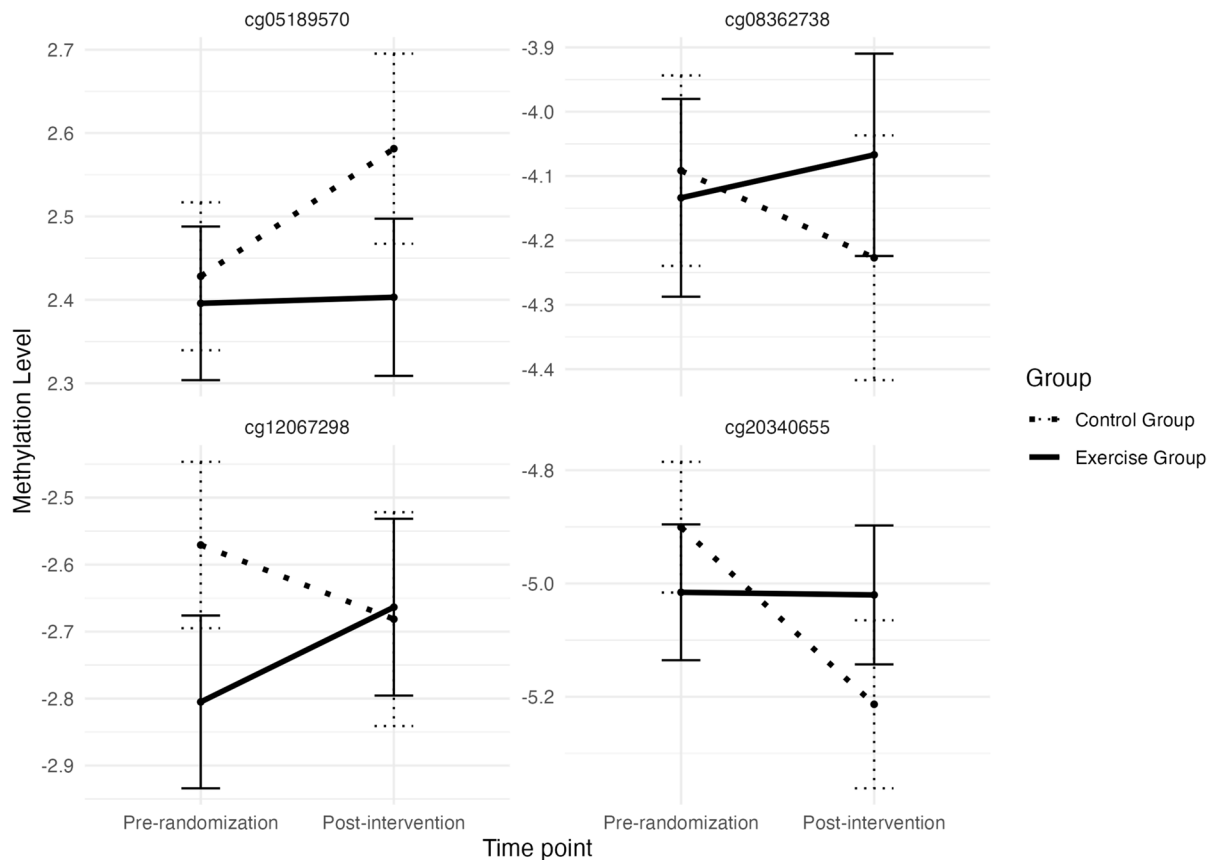
SD standard deviation, BMI Body mass index, DCIS Ductal carcinoma in situ

<sup>‡</sup>Defined as American Indian, Asian, Native American, or more than 1 race

\*Correlation is significant at the 0.05 level (two tailed)

findings should be interpreted cautiously. While the methylation status (Increased/Decreased methylation) was intensified in the usual care group, the status was either unchanged or mitigated in the exercise group. For instance, cg05189570 showed an increased methylation at pre-randomization (mean=2.413,); the level of methylation did not change in the exercise group ( $p=0.908$ ) but increased in the usual care group ( $p=0.007$ ), showing further increased methylation. Cg20340655 showed a decreased methylation at pre-randomization (mean=−4.958,); the level of methylation did not change in the exercise group ( $p=0.999$ ) but decreased in the usual care group ( $p<0.001$ ), leading to further decreased methylation. Lastly, at cg08362738

(mean=−4.110) and cg12067298 (mean=−2.684), showing decreased methylation at pre-randomization, the level of methylation increased, leading to a less decreased methylation in the exercise group, but decreased in the usual care group, leading to further decreased methylation. Significant main effects of time across the groups were detected at cg23426002, cg05847680, and cg11806762 with increased methylation at post-intervention compared to pre-randomization. Significant main effects of group in methylation, that is, group main effects averaged across the two time points, were also observed, with generally milder methylation status in the exercise group. Results were



**Fig. 2** This figure illustrates the intervention effects (exercise vs. usual care over time) on DNA methylation, measured by M-values

unchanged in sensitivity analyses restricted to White participants.

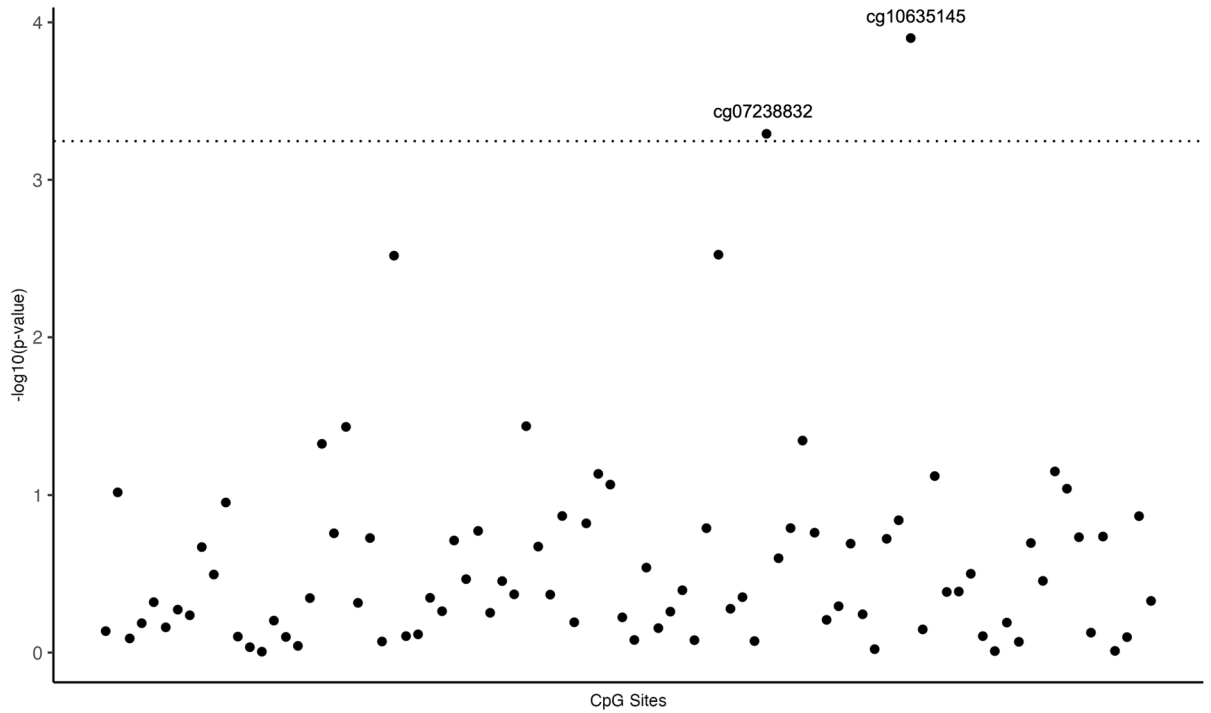
#### Association of rs6265 genotype with DNA methylation at pre-randomization

The cross-sectional meQTL analysis identified significant effects of the rs6265 genotype on DNA methylation levels at cg10635145 ( $b = -0.345$ ,  $p < 0.001$ ) and cg07238832 ( $b = -0.198$ ,  $p < 0.001$ ) (Fig. 3). Methylation levels varied by genotype, showing an additive effect of the variant T allele, with TT exhibiting the most decreased level of methylation at both sites (Fig. 4). Results were unchanged in the sensitivity analyses restricted to White participants.

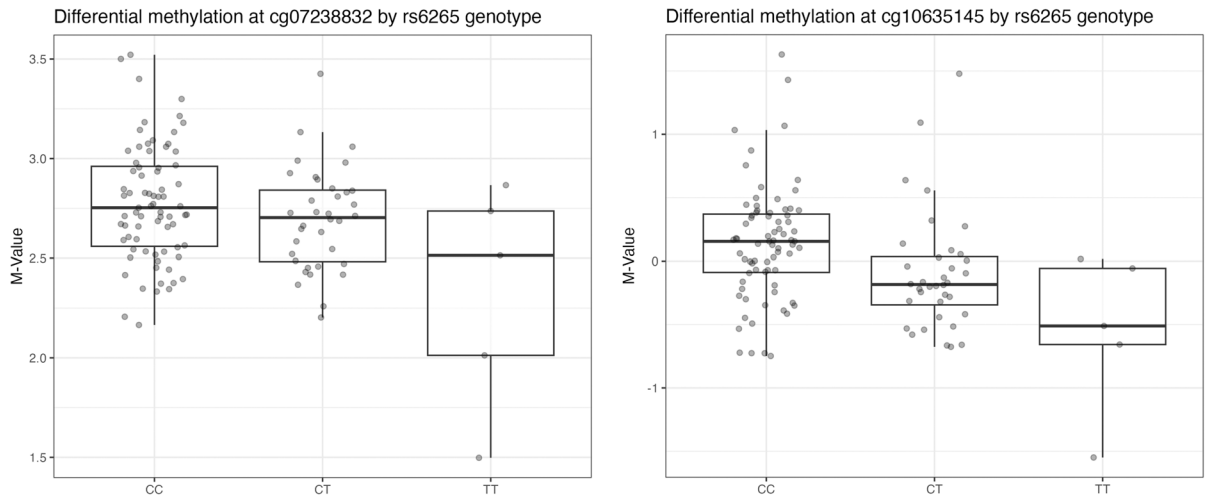
#### Association between changes in DNA methylation and changes in cognitive function

Across groups, changes in methylation at two CpG sites were associated with changes in specific cognitive domains (Table 2); cg05818894 with attention (unadjusted  $b = 0.138$ ,  $p = 0.049$ , 95%CI: 0.006 to 0.271) and cg06025631 with mental flexibility (unadjusted  $b = 0.373$ ,  $p = 0.031$ , 95%CI: 0.039 to 0.708). Model comparison based on BIC indicated a better fit for the unadjusted models relative to the adjusted models (smaller BIC values: cg05818894 = 103.62 vs. 116.64; cg06025631 = 293.35 vs. 304.37). For CpG sites showing significant intervention effects, subgroup analyses stratifying the group assignment did not find any significant differences. Results were





**Fig. 3** This figure presents the meQTL analysis between rs6265 and DNA methylation levels at CpG sites within the BDNF gene at pre-randomization. The dotted line indicates the Bonferroni-corrected significance threshold at  $-\log_{10}(0.05/88)$



**Fig. 4** This figure compares DNA methylation levels across rs6265 genotypes

unchanged in sensitivity analyses restricted to White participants.

Moderation analysis

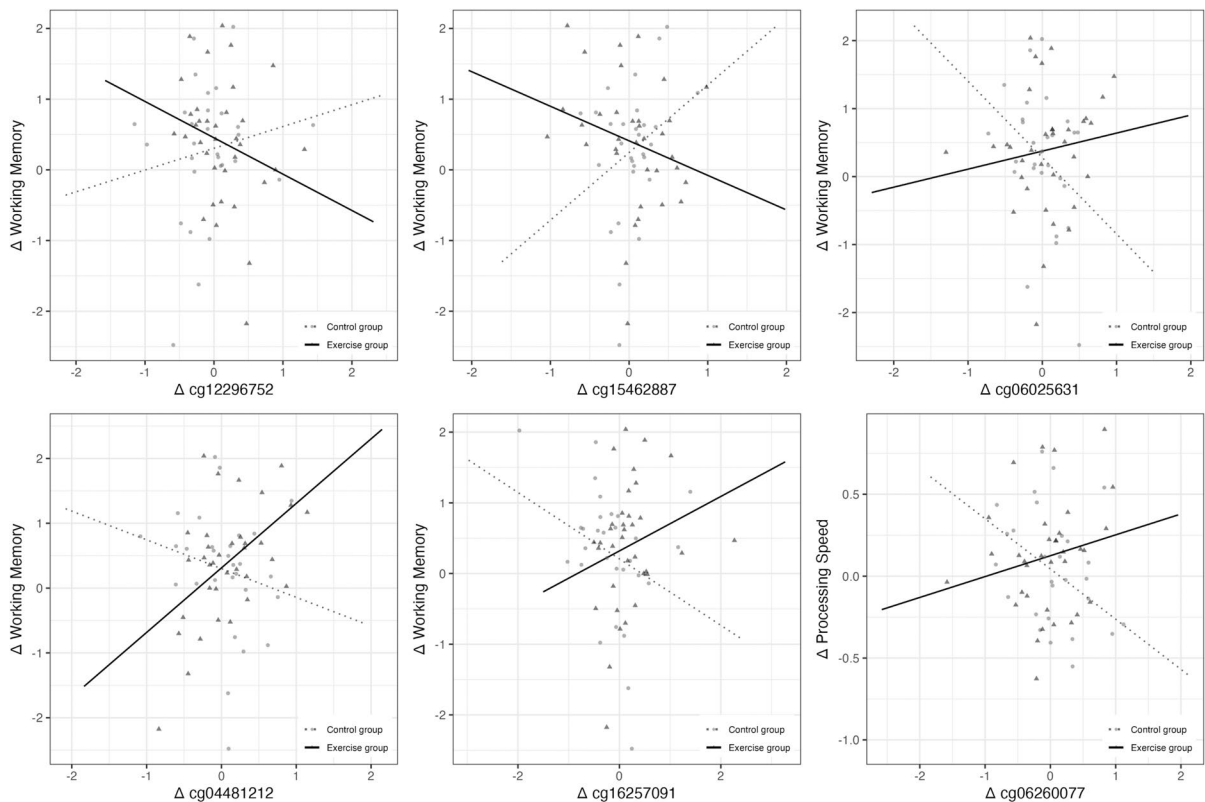
Significant interaction effects between group assignment and changes in DNA methylation were observed

**Table 2**  $\Delta$  DNA Methylation of CpG sites in the *BDNF* gene and  $\Delta$  Cognitive Function

|                             |                     | Coefficient <i>b</i> (SE) | <i>p</i> | 95% CI        | BIC    |
|-----------------------------|---------------------|---------------------------|----------|---------------|--------|
| $\Delta$ Attention          |                     |                           |          |               |        |
| Unadjusted model            | $\Delta$ cg05818894 | 0.133 (0.067)             | 0.056    | 0.002, 0.264  | 103.62 |
| Adjusted model              | $\Delta$ cg05818894 | 0.138 (0.068)             | 0.049    | 0.006, 0.271  | 116.64 |
|                             | Group               | 0.030 (0.068)             | 0.663    | −0.104, 0.164 |        |
|                             | Age                 | 0.004 (0.004)             | 0.359    | −0.005, 0.013 |        |
|                             | Education           | −0.003 (0.013)            | 0.840    | −0.029, 0.023 |        |
| $\Delta$ Mental Flexibility |                     |                           |          |               |        |
| Unadjusted model            | $\Delta$ cg06025631 | 0.365 (0.169)             | 0.033    | 0.033, 0.697  | 293.35 |
| Adjusted model              | $\Delta$ cg06025631 | 0.373 (0.171)             | 0.031    | 0.039, 0.708  | 304.37 |
|                             | Group               | −0.165 (0.151)            | 0.278    | −0.460, 0.131 |        |
|                             | Age                 | −0.001 (0.010)            | 0.918    | −0.020, 0.018 |        |
|                             | Education           | 0.043 (0.029)             | 0.149    | −0.015, 0.100 |        |

In adjusted models, Group assignment (0: Control group, 1: Exercise group), Age (year), and Education (year) were controlled

*SE* Standard Error, *CI* Confidence Interval, *BIC* Bayesian Information Criterion,  $\Delta$  Changes in...

**Fig. 5** This figure depicts the interaction effects between group assignment and changes in DNA methylation on cognitive function

at five CpG sites for working memory and one site for processing speed, indicating that the exercise effects on cognitive improvement differed by the level of methylation changes (Fig. 5, Supplementary Material 2). Methylation changes at cg12296752

( $b = -1.369$ , 95% CI:  $-2.949$  to  $-0.268$ ,  $\Delta R^2 = 0.10$ ), cg15462887 ( $b = -1.416$ , 95% CI:  $-3.035$  to  $-0.270$ ,  $\Delta R^2 = 0.07$ ), cg06025631 ( $b = 1.533$ , 95% CI:  $0.212$  to  $2.934$ ,  $\Delta R^2 = 0.08$ ), cg04481212 ( $b = 1.345$ , 95% CI:  $0.354$  to  $2.264$ ,  $\Delta R^2 = 0.10$ ), and cg16257091

( $b=0.931$ , 95% CI: 0.318 to 2.261,  $\Delta R^2=0.08$ ) moderated the effect of exercise on working memory. Specifically, participants in the exercise group showed greater working memory improvements when methylation changes were 1 SD below the mean at cg12296752 or cg15462887, and when methylation changes were 1 SD above the mean at cg06025631, cg04481212, and cg16257091. For processing speed, methylation changes at cg06260077 ( $b=0.424$ , 95% CI: 0.084 to 0.758,  $\Delta R^2=0.09$ ) moderated the exercise effects, with greater improvements observed in the exercise group when methylation changes were 1 SD above the mean.

As the overall model did not reach significance, no evidence of moderation by rs6265 genotype on the exercise effects on cognitive function was observed. Results were unchanged in the sensitivity analyses with or without participants carrying the TT genotype, and when analyses were restricted to White participants.

## Discussion

To our knowledge, this is the first study using an aerobic exercise RCT to demonstrate not only that peripheral DNA methylation of the *BDNF* gene changes, but that it has a meaningful relationship with cognitive improvements related to exercise in postmenopausal women with breast cancer. Our main findings include differential peripheral methylation at CpG-site level in the exercise group compared to the usual care group, associations between changes in peripheral DNA methylation and changes in cognitive function, and moderating effects of peripheral methylation changes on the cognitive response to exercise. Given a critical limitation of our modest a priori statistical power to detect group-by-time interaction effects and an absence of correction for multiple testing, these findings should be interpreted cautiously and viewed as hypothesis-generating, pending replication in larger confirmatory studies. Nonetheless, our findings underscore the potential of peripheral CpG-specific methylation as a key modulator of cognitive changes and highlight possible mechanisms of aerobic exercise in mitigating cognitive decline via epigenetic mechanisms.

Previous studies in rodent models have shown that short-term voluntary aerobic exercise decreases DNA

methylation at or near promoter IV in the hippocampus [36], and at exon IV and IX in the prefrontal cortex [37]. Methylation changes following exercise exhibited exon- and CpG-site-specific patterns in the prefrontal cortex, coinciding with increased *BDNF* expression [37]. Similarly, human studies have demonstrated tissue- and gene-specific DNA methylation changes following exercise [38]. These findings align with prior studies showing exercise-induced methylation changes at exons IV and IX—three located in or near exon IV and one in exon IX of the *BDNF* gene, supporting the relevance of these regions in *BDNF* regulation. The identification of CpG sites within these same exons in our analysis suggests that exercise may similarly affect *BDNF* regulation in humans through exon-specific epigenetic mechanisms. While we did not test mediation due to limited power, the location of these methylation changes within functionally important exons—exon IV, involved in activity-dependent transcription and exon IX, the sole coding exon of the *BDNF* gene [39] where the rs6265 polymorphism is located—highlights their potential biological significance for neuroplasticity and cognitive outcomes.

Direct comparison with prior studies is limited due to the lack of research in this population as well as differences in tissue types [40, 41]. To be said, methylation changes detected in plasma might not actually reflect methylation patterns or changes in brain tissue. We hypothesized that CpG-site-specific methylation changes would be observed, with smaller changes expected in the exercise group. Consistent with this, our findings suggest that aerobic exercise may play a role in maintaining epigenetic stability or attenuating aging-related epigenetic changes rather than exerting a uniform hypomethylating effect. Specifically, when the site was hypomethylated at pre-randomization, the site remained hypomethylated and it became further hypomethylated in the usual care group whereas the site methylation level became less hypomethylated or did not change in the exercise group. Similarly, if the site was hypermethylated at pre-randomization, the site remained hypermethylated and became further hypermethylated in controls whereas the site's methylation status was unchanged in the exercise group. These findings raise intriguing questions about whether exercise helps stabilize aging-related CpG-site methylation changes, potentially mitigating epigenetic age acceleration.

One possible explanation for the observed patterns is that aerobic exercise may exert a potential stabilizing influence on CpG-site methylation, thereby counteracting age-related epigenetic alterations. Aging is associated with widespread changes in DNA methylation [42], including both global hypomethylation and site-specific hypermethylation, which can influence gene expression and contribute to cognitive decline. By potentially helping to maintain epigenetic stability, aerobic exercise could preserve the regulatory functions of DNA methylation and mitigate the adverse effects of aging on cognitive function.

Another mechanism through which exercise may contribute to methylation stability involves its role in reducing inflammation, oxidative stress, and other aging-accelerating environmental factors. Chronic inflammation and increased oxidative stress have been implicated in both neurodegeneration and epigenetic dysregulation, leading to accelerated cognitive decline [43]. Exercise has been shown to modulate inflammatory cytokine levels and enhance the body's antioxidant defenses, which may, in turn, influence methylation patterns and maintain neuronal integrity [44].

Cellular aging has been associated with cognitive decline [45–47], providing a relevant context for interpreting exercise-related effects on genomic stability and epigenetic regulation. Aerobic exercise has also been linked to slower progression of biological aging as measured by epigenetic clocks [48], which may represent a potential pathway through which exercise supports cognitive health. By slowing epigenetic aging, aerobic exercise may help preserve cognitive function and reduce the risk of age-related neurodegeneration.

Taken together, these findings suggest that exercise-induced epigenetic modifications may serve as a critical mechanism linking exercise to cognitive function. Further research is needed to determine whether these effects translate into long-term cognitive benefits and whether targeted exercise interventions can optimize epigenetic aging trajectories.

Although our analysis was conducted in a cancer population, CpG sites showing exercise-related effects have previously been implicated in neurodegenerative and stress-related conditions. For instance, cg08362783 and cg20340655 showed decreased methylation in individuals with presymptomatic dementia [49] and posttraumatic stress syndrome

[50] respectively. Methylation at cg20340655 was even lower in a high trauma group compared to a low trauma group [50]. Maternal exposure to war trauma was associated with increased methylation at cg05189570 [51]. In our analysis, cg12067298 was newly identified as a differentially methylated site following exercise.

Our meQTL analysis detected two CpG sites exhibiting differential methylation by rs6265 genotype, with an additive effect of the T allele leading to lower methylation. This result suggests a potential regulatory role of rs6265 in modulating peripheral DNA methylation, where the T allele carriers may exhibit site-specific methylation profiles affecting gene expression and neuroplasticity. Taken together, these results point to a dose-dependent epigenetic influence of rs6265, although further validation is required to confirm these associations.

Interestingly, despite prior evidence suggesting that the CC genotype may confer greater cognitive benefits from exercise in healthy adults [27, 52], we did not observe a significant moderating effect of rs6265 on cognitive improvements following exercise, consistently in the sensitivity analyses. This may reflect limited statistical power, as the T allele, particularly in the homozygous state, occurred at low frequency in our cohort. This could also be attributed to the lower prevalence of the T allele in European populations, which is the predominant ethnicity of our sample [53]. However, given that our sample's genotype distribution was consistent with Hardy–Weinberg equilibrium and aligned with the expected proportions in other European-descent populations, the lack of significance is unlikely to be due to sampling bias alone. Other factors, such as environmental influences or limited statistical power, may have contributed. Moreover, prior studies have linked the T allele to an increased risk of Alzheimer's disease in female European descents, relative risk of 1.18 [54], and greater susceptibility to cancer-related cognitive decline [55], whereas opposite trends have been observed in Chinese populations [56, 57]. Our findings provide insights into genetic influences on cognitive aging with genotype and population-specific patterns.

Our findings also suggest that peripheral DNA methylation may play a more dynamic role than genetic variation alone in shaping cognitive responses to exercise, consistent with the concept of epigenetic

plasticity. We identified CpG sites whose peripheral methylation changes moderated cognitive improvements, particularly in working memory—the cognitive domain most frequently moderated by peripheral methylation changes. Working memory is highly sensitive to neurobiological changes, involving prefrontal, parietal, and temporal cortical networks [58–60]. Prior studies suggest that working memory is especially adaptable to exercise-induced improvements via neuroplasticity [61–63].

Among the CpG sites showing moderating effects, cg04481212 was previously linked to trauma-related epigenetic modifications. Vietnam veterans with trauma exposure exhibited decreased methylation at this site, possibly reflecting long-term biological adaptations to stress [64]. In our analysis, greater methylation decreases at cg04481212 were associated with greater cognitive improvements following aerobic exercise on working memory, suggesting that epigenetic modifications may enhance brain responsiveness to exercise.

No overlap was observed between the CpG sites responsive to exercise and those with changes associated with cognitive changes. Of note, cg06025631, which did not show an intervention effect, was associated directly with mental flexibility and conditionally with working memory. This pattern raises the possibility that certain loci may serve more as baseline or moderating markers rather than direct targets of exercise-induced change.

A few findings warrant particular attention given their somewhat counterintuitive directions, exemplified by cg05818894 and cg06025631. cg05818894 is in the *BDNF* gene body (between exon 3 and exon 4), where methylation is often linked to transcriptional activation rather than repression [65], which may help explain its positive association with improvements in attention. By contrast, cg06025631 lies within the *BDNF* promoter, where hypermethylation is typically repressive [65]. Its associations with two domains could therefore suggest a pleiotropic influence of changes in peripheral methylation on cognitive processes [66], consistent with the central role of *BDNF* in supporting both prefrontal and hippocampal circuits underlying executive functioning and memory processes [15]. Alternatively, given its promoter location, hypermethylation at this locus could reflect context-dependent regulation, silencing of a repressive element [65], whereby baseline peripheral

methylation at this locus influences intrinsic cognitive capacity while simultaneously shaping responsiveness to exercise. These site-specific observations underscore the complexity of *BDNF* regulation and, given the exploratory design and multiple testing, should be viewed as hypothesis-generating pending replication in larger samples.

The identification of CpG sites significantly associated with cognitive improvements in response to aerobic exercise suggests a potential causal role of DNA methylation. Although causality cannot be inferred from our analysis, the reversibility of DNA methylation raises the possibility of compensatory mechanisms aimed at maintaining cognitive function, particularly in individuals susceptible to decline [67]. Different CpG sites may exert distinct effects, with some conferring cognitive resilience while others indicating cognitive vulnerability—possibly through neuroinflammatory pathways, synaptic integrity, or neurodegenerative processes.

Given that DNA methylation in peripheral blood may not fully reflect brain-specific changes, future research should investigate whether these epigenetic modifications correlate with neurocognitive function in brain regions such as the prefrontal cortex and hippocampus. Integrating neuroimaging and proteomics could enhance mechanistic understanding and validate peripheral methylation biomarkers.

Certain limitations should be considered. Because allele frequencies of rs6265 and DNA methylation patterns vary by ancestry, the predominantly White composition of our sample may limit the generalizability of our findings to other racial and ethnic groups. Nonetheless, sensitivity analyses restricted to White participants produced results consistent with the full sample, supporting the robustness of our conclusions. Despite leveraging data from an existing RCT with adequate power for exploratory analyses, the modest sample size limited our ability to detect small intervention effects. This constrains the generalizability of our findings and underscores the need for replication in larger and more diverse cohorts to confirm observed patterns and evaluate more subtle intervention effects. In addition, the six-month timeframe may not fully capture long-term methylation changes, and although we observed associations between methylation and cognitive function, causal inference remains limited. Moreover, better model

fit in a few unadjusted models suggests the presence of potential confounding factors. Future research should investigate longitudinal methylation trajectories across diverse time intervals, incorporate multi-omics and neuroimaging approaches, and examine the interplay between genetic, epigenetic, and environmental influences to better elucidate the underlying molecular mechanisms of cancer-related cognitive decline accompanying accelerated aging.

A key strength of this study is its longitudinal design, allowing us to track peripheral DNA methylation and cognitive trajectories over time and especially in response to aerobic exercise. The use of an RCT design, rigorous statistical analyses, and a battery of objective cognitive function measures enhances the robustness of our findings. Notably, the exceptionally high adherence (99.8%) to the six-month intervention strengthens confidence in our results. Furthermore, the reversible nature of DNA methylation, along with its variability across individuals and tissues [68], enhances the translational relevance of our findings. This knowledge may support the identification of predictive and prognostic biomarkers of cognitive aging, facilitating earlier detection of individuals at greater risk of decline or those more likely to benefit cognitively from exercise interventions.

Overall, this analysis provides preliminary evidence that aerobic exercise may influence cognitive function via epigenetic mechanisms, especially CpG-site specific DNA methylation. These findings highlight methylation as a potentially modifiable biomarker for cognitive aging and emphasize the potential for precision interventions that promote cognitive resilience in aging populations.

**Acknowledgements** The authors gratefully acknowledge all participants and research staff of the EPICC trial.

**Author contribution** The study was conceptualized by MC, CB, YC, SS, and KI. MC and CB were responsible for the methodological design and MC and SS were responsible for the data analysis. Data curation was performed by MC. The initial draft of the manuscript was written by MC, with all authors contributing to the review and editing process. CB supervised the project. All authors have read and approved the final manuscript.

**Funding** This research was supported by the National Cancer Institute grants (R01-CA196762 and R01-CA221882), the University of Pittsburgh School of Nursing grants (E. Jane Martin Research Award and Janice Scully Dorman Endowed Omics

Research Fund), and the National Institute of Nursing Research (T32-NR009759).

**Data availability** The datasets generated during this study are subject to controlled access due to participant privacy. They are available from the corresponding author upon reasonable request and with the approval of the dbGaP Data Access Committee (accession number: phs003959.v1.p1).

## Declarations

**Conflict of interest** The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

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